Independent review of the first scattering-paper draft

Review for PI Gustav Delius

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Recommendation: revise before circulation. (d) The core classical phase survives the independent checks **(c)**. The nine appendix items do not invalidate the computed classical scattering phase, but two have consequences beyond a local typo: the bad cubic sign reappears in a normal-ordering energy claim, and the signed defect-root dictionary misidentifies the physical defect for negative orientation **(c)**. There are **14 required corrections**, including coordinate and charge definitions and qualifications of stability and fusion. The corrected good classical energy was not used incorrectly in the inspected one-loop calculations **(c)**. Passing algebraic checks does not settle the physical one-loop determinant or quantum frame **(d)**.

Scope and provenance. This review reads `AGENTS.md`, `README.md`, the full draft, the five requested source write-ups/map, the four named one-loop/charge write-ups, relevant downstream notes, and the cited script implementations. Only the three requested deliverables are changed; no git operation is used. Labels retain their repository meanings. Review findings, numerical counterexamples and source/code observations are **(c)**; recommendations and unestablished physical extensions are **(d)**. All new mathematical calculations below are checked in `paper/scripts/review_draft1_checks.py`; the run groups are named A1–A9 and B. Source-script evidence is distinguished from an independent proof.

The original bibliography's reading labels belong to earlier project agents. For this review, the Corrigan–Zambon full-text sections 2–5 were read online, especially (3.22), (3.37), (3.43)–(3.44), and (5.5). The Bogdan–Charkina full text was consulted at the classical energy, breather and action formulas; not every page was reread. The Charlier–Lenells water-wave abstract/entry was independently verified; the large HTML full text could not be retrieved. Other literature claims are assessed against the supplied literature map and source write-ups, not certified by a fresh full-text literature review. No new reference is invented. The precise reading limitations matter to corrections R12–R13.

A. Triage of all nine appendix items

Item	Correct version **(c)**	Evidence	Downstream effect **(c)**
1. Bad cubic	$\mathcal{L}_b = \frac{1}{2}\phi_t^2 - \frac{1}{2}\phi_x^2 - \phi_x^3 + \frac{1}{2}\phi_{xx}^2$. $E_b = \frac{2}{15}k^3(5+4k^2)$.	A1: exact Euler–Lagrange calculation; exact pulse integrals and 65-digit direct quadrature.	Classical phase, M, Π, P and the computed energy use the correct data. However, `phase2/oneloop.tex` and the docstring of `oneloop_soliton_energy.py` give the opposite sign for the normal-ordering boundary energy. Their displayed numerical tables use scheme term zero; those tables are unaffected. See A.1 below.

Item	Correct version **(c)**	Evidence	Downstream effect **(c)**
2. Good energy	$E_g = \frac{2}{15}k^3(5-4k^2)$; $dE_g = v\,dP$. The printed one-loop energy exceeds this by $k^3(5-2k^2)/15$.	A2: symbolic integration, direct field-energy quadrature for both directions; actual downstream function bodies evaluated independently.	None found in the specified one-loop renormalisation, charge or phase calculations. `audit_charges_action.py:ecl` explicitly uses the correct energy. The incorrect displayed energy and the adjacent asserted derivative identity must both be corrected. This does not clear the separate physical one-loop audits.
3. Defect root	At fixed CZ fields and orientation: $s = \sigma_0$, $q = -s/(2\sqrt{3})$, $\alpha_{CZ} = \eta/\sqrt{3}$, $\beta_{CZ} = 2\eta/\sqrt{3}$. The spectrum may separately use $\eta\sigma_{vac}$.	A3: CZ constraints and nonlinear sewing on a one-soliton for both η and both directions. A signed-root substitution at fixed defect fails.	Wrong physical root labels propagate into the text of `oneloop_defect_phase.tex` and Route E of `report.tex`. The two `oneloop_defect_*` calculation scripts use the unsigned root. Their functions and formal chord identities are unaffected; interpretation as functions of the same CZ σ for $\eta = -1$ is affected.
4. Transmission	$z = ((v+s-\sqrt{3}k)/(v+s+\sqrt{3}k))^\eta = ((p_b-q)/(p_a-q))^\eta$. The printed middle expression is $-1/z$.	A4: rational identity for both orientations; comparison with actual `z_defect_mode`; nonlinear sewing.	None found in executable transmission, delay, phase or fusion formulas. They use the rational/root expression, not the incorrect middle expression. It is a real sign and reciprocal error, not a harmless logarithm branch.
5. KdV ratio	The *full* derivative ratio is $0.564246498175\ldots$ at ratio 1.7; the cross-only ratio is $-0.435753501825\ldots$.	A5: independent high-precision small-amplitude derivative with scale 10^{-12}; asymptotic expression below.	The master-report prose is wrong, and the final explanatory print line of `consistency_checks.py` has a minus-sign typo. Its computed numerical ratio and the paper's precise ratio are correct. No phase value changes.
6. Script names	`smatrix_poles_bootstrap.py`, `toda_quantum_corrections.py` are the existing names.	A6: filesystem inventory; old basenames absent, replacements present under `phase2/scripts`.	Documentation/reproducibility only. The two omitted S-matrix checks exist. All explicitly cited script paths in paper Sections 1–6 exist.
7. Root notation	Ordered $\mu_1 = -p_b$, $\mu_2 = -p_a$; $\lambda = -\ell/(12\sqrt{3})$. Bad roots instead obey $\mu^3 + \mu/4 = i\ell/(12\sqrt{3})$.	A7 and paper conventions check 4; roots substituted in the two cubics and cross-ratio checked in both real forms.	None for this notation change. It requires an ordered dictionary, not just an unidentified statement "$\mu = -p$". Additional bad transcription formulas in Phase 2g are noted below; the paper does not copy them.
8. Toda width	$\kappa_H = 2\kappa_T$, and $\kappa_H \sim \sqrt{12}\,k_b$, $\kappa_T \sim \sqrt{3}\,\beta$ at small amplitude.	A8: exact factor two in the exponential and $\sinh\beta/\beta \to 1$; source lattice formulas and scaling checks.	None. These are asymptotic identifications; $k_b = \sinh\beta$ must not be replaced by β at finite amplitude.
9. Phase number	`defect_semiclassics.pdf` is Phase 2c.	A9: title, README and source inventory.	None. Phase 2d in the one-loop defect narrative is a citation typo.

A.1 Energy signs and the downstream trace

For $u = k^2 \operatorname{sech}^2(kx/2)/2$ and $\pi = -vu$, direct integration gives **(c)** (A1–A2)

$$\int u^2 dx = \frac{2}{3}k^3, \qquad \int u^3 dx = \frac{4}{15}k^5, \qquad \int u_x^2 dx = \frac{2}{15}k^5.$$

Inserting these in the two Hamiltonians gives the energy formulas in the table. With the wrong bad cubic sign, the energy on the same positive pulse would be $2k^3/3$, differing from the correct value by $-8k^5/15$; moreover that pulse is not a solution of that wrong equation **(c)** (A1). The wrong good energy has derivative residual **(c)** (A2)

$$\frac{dE_{\text{printed}}}{dk} - v\frac{dP}{dk} = k^2\left(1 - \frac{2}{3}k^2\right) \neq 0 \quad (0 < k < 1).$$

This form avoids division by P' at the cusp.

The bad Hamiltonian contains $+u^3$. Under the usual Wick replacement $u^3 \mapsto {:}u^3{:} = u^3 - 3Cu$, its energy shift is **(c)** (A1)

$$\Delta E_{\text{Wick}} = -3CM = -\frac{3\hbar}{2\pi}M = -\frac{3\hbar}{\pi}\sinh\beta, \qquad C = \frac{\hbar}{2\pi}$$

for the previously selected stable-band vacuum. The positive shift asserted in `phase2/oneloop.tex`, the paragraph "Decision (PI)" and the energy table caption, is the sign appropriate to the *good* Hamiltonian. The same positive claim occurs at lines 15–16 of `phase2/scripts/oneloop_soliton_energy.py`. Its runtime code only computes C and the zero-scheme integral; it does not add either boundary term. `oneloop_breather.py` also uses the zero-scheme values. Thus this is propagation into a claimed renormalised energy, not evidence that the numerical zero-scheme tables or classical phase were computed with the wrong Lagrangian **(c)**. No new scheme is selected by this review.

For the good-energy typo, the dependency trace is:

- `oneloop_good_phonon_scattering.py` builds modes from the PDE and Hirota data, with no classical energy input. **(c)**
- `oneloop_good_twosoliton_phase.py` computes the continuum integrals, `Delta_P1_closed`, `Delta_E1_ren_closed`, and `delta_renorm_good`. The last uses derivatives of the classical phase and momentum. Independent checks reproduce it in both collision channels. `oneloop_good_limits_cusp_resonance.py` imports those quantities. **(c)**
- `charge_and_frame_audit.py` computes $L_1 = v\Delta P_{mode} - \Delta E_1$ from mode sums and differentiates it. The new script evaluates these exact function bodies and verifies their agreement with the preceding mode charges. No E_{printed} is used. **(c)**
- `audit_charges_action.py:ecl` is $2\sin^3\alpha(1+4\cos^2\alpha)/15$, the correct classical energy. Its finite-dimensional charge argument and the corresponding write-up do not inherit the typo. **(c)**

The remaining charge/frame and determinant limitations in the independent audits are separate issues; correcting E_{printed} does not resolve them **(d)**.

A.2 The defect convention is settled, not an open choice

CZ's full-text equations impose $\beta_{CZ} = 2\alpha_{CZ}$ and $\beta_{CZ}^2 - \alpha_{CZ}^2 = 1$ in the stated units **(a)**. The later pair $\alpha_{CZ} = 1/(2\sqrt{3})$, $\beta_{CZ} = \eta/(2\sqrt{3})$ violates both constraints **(c)** (A3). It is not an alternative parametrisation of that same action. The source's s in (5.5) is σ_0 when $c = 1$. Corrigan–Zambon, full-text sections 3 and 5.

For example, $\alpha = 0.4$, $\epsilon = +1$, $\sigma_0 = 2.4$, $\eta = -1$ gives $z = 1.50971083399\ldots$. Replacing the root by $-\eta\sigma_0/(2\sqrt{3})$ gives $z = 0.373565030471\ldots$; the maximum residual of the four sewing equations at $t = 0.27, x = 0$ is $3.3312537\ldots$, while the correct factor passes below 10^{-52} **(c)**

(A3). Thus transmission, delay, absorption-root identification and root-labelled phase/charge values change if the later dictionary is applied literally. The spectrum legitimately depends on $\eta\sigma_{vac}$, but that combination must not replace σ_{vac} in the scattering root without a simultaneous parity transformation of the bulk data **(c)**.

Both `oneloop_defect_scattering_and_stability.py` and `oneloop_defect_phase_and_fusion.py` retain $s = -2\sqrt{3}q$ and the correct rational/root factor; their executable functions do not use the incorrect middle expression. The later master report copies the bad dictionary in its Route E prose. The mirrored-triple loop in `audit_frames_defects.py` uses an extra η in its local variable `sigmas`; it still counts one stable root, since both orientations and reflected triples are sampled, but does not establish equality of the two physical root dictionaries **(c)**. Its determinant checks use the signed spectral parameter legitimately.

Two nearby Phase 2g transcription errors should not be imported when repairing the dictionary: its displayed cubic has $\ell/(4\sqrt{3})$ in place of $\ell/(12\sqrt{3})$, and its trigonometric expressions for p_a, p_b do not equal their preceding Cartesian expressions. The Cartesian roots used by the code and by the paper are correct **(c)** (A7/B root substitutions).

A.3 Minor-item precision

The independent KdV limit is **(c)** (A5)

$$\frac{\partial_{\beta_1}\delta}{2\beta_1 \log A} \longrightarrow 1 + \frac{2\beta_2}{\beta_1 \log A_{KdV}}, \qquad A_{KdV} = \left(\frac{\beta_1 - \beta_2}{\beta_1 + \beta_2}\right)^2.$$

The extra $4\beta_2$ is the cross-term contribution to the *phase derivative*, namely $-\Pi_1' M_2$, not the dimensionally different spatial shift itself. The paper mostly repairs the earlier prose, but its small amplitude paragraph should use this precise name. Script inventories are filesystem observations, not mathematical tests; A6 and A9 test their stated existence/title assertions rather than fabricating symbolic checks.

B. Required corrections to the paper

Locations refer to source labels and subsection names, so they remain usable if another agent changes line numbers. Each item states a review finding **(c)** and a recommended fix **(d)**. The list contains exactly 14 items.

R1. **Proposition `prop:EP`, one-soliton coordinate.** Equation `eq:1sol` defines x_0 as the additive exponent, not the centre. At $t = 0$, $X = -x_0/k$ and the pullback is $\Omega = dP \wedge dX = -P_k\, dk \wedge dx_0/k$. Replace the printed $\pm P_k\, dk \wedge dx_0$ by this formula, or explicitly rename the physical centre X. Direct quadrature gives -0.78 at $k = 0.6$, where $P_k = 0.468$ (B). The degeneracy at the cusp survives. The later phase scripts use physical positions and do not inherit this normalisation error.

R2. **Section 2, Gelfand–Dickey charges.** The displayed definition $H_n = \int \mathrm{Res}\, L^{n/3} dx$ is divergent on the nonzero Lax background. Already $\mathrm{Res}\, L^{1/3} \to -1/12$ (B). Use the vacuum-subtracted residue, as `charges.py:H` actually does. The quoted finite H_1, H_2, H_4, H_5 and their E, P combinations then agree with the code; label this definition explicitly before using them.

R3. **Section 2, paragraph after `eq:shifts`, and Section 4 small amplitudes.** The KdV expression printed as a position shift is $\log A_{KdV}$. For the stated ordering, write $\Delta x_1 \sim -2k_1^{-1} \log|(k_1 - k_2)/(k_1 + k_2)|$ and $\Delta x_2 \sim +2k_2^{-1} \log|(k_1 - k_2)/(k_1 + k_2)|$. Also call $-\Pi_1' M_2 \to 4\beta_2$ a contribution to the phase derivative, as distinguished in A.3. The exact shifts and phase are unaffected.

R4. **Section 4, Toda set-up.** Restore the direction sign in $dE/dP_n = sb\sinh\kappa/\kappa$. The unsigned expression is false for left-movers (B). It is inherited from `toda_route.tex`; the executable Toda phase functions use the signed velocity, so the head-on scaling test survives.

R5. **Section 5, defect symplectic structure.** With the stated canonical one-form and $\Omega = d\Theta$, the contribution is $(g_u - f_v)du\wedge dv$, not $(f_v - g_u)du\wedge dv$. Change that sign or explain a changed convention. Its restriction to a fixed vacuum sector is still zero, so no soliton–defect cross term is introduced. Do not infer absence of localized defect energy or momentum from this vanishing two-form; the paper already correctly calls M_D, Π_D effective phase coefficients.

R6. **Abstract, summary, `prop:stab`, and the ensuing consequences.** Qualify the stability theorem by its evidence: the cited finite root search is not a proof of absence of all unstable modes. State the sampled result as **(c)** and the unsupplied completeness assertion as **(d)**, or provide an admissible-root/completeness proof. Separate the marginal case $\eta\sigma = -1$ from strict stability. Restrict "one stable vacuum per triple" to three distinct real roots. For example $s = 2$ has vacua $2, -1, -1$; $|s| > 2$ has only one real vacuum (B). The unrestricted statement "both excited vacua are unstable" is likewise inapplicable when those vacua are nonreal. These qualifications are already recognised in `audit_frames_defects.tex`.

R7. **Proposition `prop:fusion`, absolute constant.** Replace "with no free constant" by "for the displayed zero-constant representatives". A shift $\delta_D^{(q)} \mapsto \delta_D^{(q)} + C(q)$ leaves every classical delay invariant and adds $C(q') - C(q)$ to the fusion identity. The cited script sets those constants to zero; it does not determine an absolute physical phase. Preserve the algebraic identity as **(c)**. Give the orientation sign explicitly, e.g. $\rho = \epsilon_3\eta\,\mathrm{sgn}(q - q')$ for the ordered chord convention used by the scripts, instead of leaving it undetermined.

R8. **"Labels and verification", verification footnotes and appendix.** Replace the assertion that every **(c)** formula already has a matching asserted test by a precise coverage statement. In particular: `charges.py` does not check the x_0 symplectic formula; `defect_two_soliton_symplectic.py`, check 5, prints limits and a *defect* cusp but does not assert the paper's two-body square-root claim; and the Toda two-body reflection amplitude is explicitly "not separately checked" in the appendix. The present B checks supply a Bessel-wave test for that last amplitude. The Hirota exact-arithmetic test covers selected rational amplitudes in every direction pattern for $N \le 4$, not a symbolic proof at all amplitudes; one separate profile assertion in that script is disabled by `orTrue`. State the actual scope, and do not count printed diagnostics as assertions. The legacy `phase0/checks/convention_check.py` also lacks the required final success marker, although its sign assertions pass.

R9. **Appendix "Inconsistencies found", all nine items.** Replace unresolved guesses with table A's verdicts and downstream trace. In particular disclose the bad normal-ordering sign, the verified CZ parameter constraints, and the negative-orientation physical mismatch; remove "could not check" and "whether it propagated ... was not checked". Retain the distinction between text errors and unaffected executable calculations. Record the minor final-print sign error in `consistency_checks.py` and qualify the Toda width map as asymptotic.

R10. **Sections 2 and 4, source dictionaries.** The paper's own good/bad dictionary and the CZ factor-nine dictionary are correct, but the imported BC and CL formulas also need explicit field maps. For BC use $\varphi_{BC} = -\phi_{bad}$ and $w_{BC} = -u_{bad}$; this leaves the canonical action/energy normalisation unchanged. For the CL bad PDE use $u_{CL} = 3u_{bad}$; for the BD field normalisation used in Phase 1, $u_{BD} = u_{good}/2$. Identify the bad roots called p in `eq:crossratio` with the preceding $\mu_{a,b}$. Otherwise the "both equations" cross-ratio uses an undefined bad-equation p. These are notation/attribution repairs, not changes to the verified continuation of δ.

R11. **Section 1, quantum prior work, affine rank.** For $a_n^{(1)}$ the underlying finite algebra is $\mathfrak{sl}_{n+1}$, not $\mathfrak{sl}_n$. Write $U_q(\widehat{\mathfrak{sl}}_{n+1})$, or use the consistent alternative notation $a_{N-1}^{(1)}$ with $\mathfrak{sl}_N$. The Section 6 $a_2/\mathfrak{sl}_3$ formulas themselves are consistent.

R12. **Section 6, obstructions; Section 1, no-boost inference.** The observation that the displayed affine-Toda and Boussinesq constants have different mass/dispersion terms does not prove that *no* limit of fixed-species affine Toda can produce Boussinesq. No cited script proves that universal exclusion. Replace it by the supported statement that no such dictionary/limit has been constructed here (**(d)** for a conjectured obstruction). Likewise absence of a boost is not alone a proof against every possible reparametrised difference form. State the checked nonseparability in the chosen rapidities, or supply the stronger proof. Describe the QISM issue as an obstruction to the ordinary ultralocal construction, not to every possible lattice quantisation.

R13. **Literature labels in Sections 1, 2 and 4.** Separate the literature's explicit classical action calculation (**(a)**) from its semiclassical Bohr–Sommerfeld prescription (**(b)**), and keep the project's string identity conditional on that prescription (**(c)**). The source expressly assumes large quantum number; it does not prove an exact quantum spectrum for every N. Specify hypotheses/function spaces when paraphrasing the CL existence theorem and the abstract-only Bona–Sachs claim. Do not present the literature-search absence result as a theorem of nonexistence. Verify the bibliography entries explicitly marked "not machine-verified" before making the blanket assertion that all references are verified. Historical reading-status labels may stay if clearly attributed to the earlier agents, as the draft's last paragraph does.

R14. **Summary item 6 and claims-table equation tags.** Split the combined "good & bad" summary into what was actually checked: the canonical non-ultralocal bracket is displayed for good, the interaction-block factorisation is checked in both forms, and the cited product-form exclusion script tests bad overtaking. Either supply the additional good exclusion check or narrow that tag. The hierarchy-charge row should also distinguish the good H_n evaluation from the two-form root dictionary. Toda itself is a lattice model; "bad" denotes its small-amplitude continuum target, not an identity of finite-amplitude theories.

Displayed-equation and claim-label ledger (Sections 1–6)

Every displayed environment in the requested sections is covered below; inline formulas with a material defect are included in the corrections. "Pass" means source agreement plus the specified check, not a claim of independent proof for every continuum or literature assertion. All audit assessments in this ledger are **(c)**.

Location / displayed formula	Source and check	Verdict / label scope
Section 1 (no displayed equation environments in its body)	Literature map; Phase 1 introduction; master-report scope.	Prior-work (a)/(b) mostly follows the map; R11–R14 qualify rank, inference and equation attribution. Abstract phase agrees with Section 4.
`eq:good`, `eq:bad`, `eq:Lag`, `eq:Omega0`, `eq:charges`	Phase 1 equations/canonical definitions; paper check 1; independent A1–A2.	Pass. The correct bad Lagrangian is the paper's improvement over its source. Potential EL equation is differentiated once to obtain the field PDE.
`eq:1sol`, `eq:rapidity`, `eq:solcharges`, `eq:dP`	Phase 1; Hirota/charges scripts; paper checks 2,3,9; independent A2.	Pass; interpret dE/dP by continuation at the cusp. Inline symplectic coordinate needs R1.

Location / displayed formula	Source and check	Verdict / label scope
`eq:roots`, `eq:cubic`; inline bad cubic and H_n	Phase 1 charges, Phase 2c transmission and bad ansatz; paper check 4; independent B.	Root equations pass. The H_n values pass after R2 subtraction. The $\mu = -p$ relation reverses ordering.
`eq:Nsol`	Phase 1/Hirota. Exact rational-amplitude checks through $N = 4$.	Agrees with source. General N is literature (a), sampled project check (c). Do not call a finite rational grid a general proof; R8.
`eq:Agood`, `eq:Abad`, `eq:crossratio`	Phase 1 interaction regimes; consistency check 5; independent B.	Pass in both forms and orientations, with bad roots named explicitly (R10). Exclude coincident solitons and singular denominators when using ordinary two-body scattering.
`eq:shifts`; inline breather constant	Phase 1 peak tracking and breather scan; paper conventions.	Exact shifts pass for soliton 1 initially left and $v_1 > v_2$. Good breather bound is only for the stated Hirota zero-background class. The following KdV sentence needs R3.
Dictionary table; unnumbered continuation display	Phase 1 Lax script; paper check 5.	Pass as an analytic identity of formulas, not a map between real solutions. CZ exclusion is only within their nonlinear type-I ansatz.
`eq:Omega`, `eq:sympid`	Master report; symplectic scattering and two-soliton defect scripts.	Pass as asymptotic pullbacks, with fixed global potential constant and well-separated solitons. Exponentially small terms are explicitly excluded.
`eq:darboux`, `eq:darbouxshifts`	Master report; semiclassical/consistency scripts; independent B.	Pass locally off $P' = 0$. In–out difference uses the ordering change. Normalising the opposite potential endpoint gives the same differences.
`eq:delta`	Master report; independent high-precision gradients, both equations and all four sign combinations.	Pass as a classical generating function in the specified frame. An exact quantum S-matrix is not established by this identity.
Inline properties of `prop:props`	Paper checks 6–7 and existing defect script check 5.	Antisymmetry and regular-branch crossing pass. Resonance finiteness follows from the vanishing spectral difference; cusp citation does not assert its two-body local expansion (R8). Logarithm sheet and regular region matter.
`eq:Ibr`, `eq:string`	BC action formula; consistency check 4; paper check 8.	Classical action identity passes. Quantisation is semiclassical and conditional (R13); exact pole positions remain (d).
Toda unnumbered M, Π, E display; inline P_n, A_T	`toda_route.tex`, lattice-soliton and label scripts.	Pass; derivative velocity needs R4 sign. Vacuum energy and pressure-work subtractions are already stated in the paper.
Toda inline R, Θ and classical limit	Route D and quantum BA; independent Bessel/Schrodinger test B.	Reflection amplitude passes with incoming $e^{-iqx/\hbar}$ and $\Delta p = 2q$. Choose an unwrapped logarithm for the classical limit. The BA method remains labelled (b).

Location / displayed formula	Source and check	Verdict / label scope
`prop:todaBA`, `eq:deltaT`, unnumbered scaling ratio	Classical BA, dressed-phase, frame, label and scaling scripts.	Matches the source and numerical checks. Mixed-derivative primes refer to the respective κ_j; not arbitrary physical-momentum derivatives. Numerical agreement is not a complete analytic convergence proof.
`eq:zCZ`, `eq:zroot`	CZ (5.5), Phase 2c transmission; independent A3/A4 sewing.	Pass with $s = \sigma_0$. Require finite positive z for ordinary transmission; the endpoints are absorption limits.
Inline Ω_D and `eq:dD`	Phase 2c transmission checks 4 and 6; exact one-form calculation.	Phase and gradient pass. R5 corrects the unrestricted two-form sign; its vacuum restriction remains zero. s-dependent constants are unfixed.
`prop:2sD`; vacuum root/absorption identities	Phase 2c one- and two-soliton scripts.	Pass for the regular sampled configurations; algebraic factorisation is consistent with the preserved A_{12}. Excited-state interpretation needs the real-vacuum restrictions in R6.
`eq:fusion`	`defect_smatrix_good_continuation.py`, check 4.	Pass for zero-constant representatives; R7. The script determines the relative orientation sign at sampled triangles, not a quantum bootstrap.
Section 6 unnumbered Poisson brackets	`routes_BC_checks.py`; independent smeared bracket B.	Pass. $\{r, r\} = 0$ follows by integration by parts for decaying test functions. The nonzero derivative-delta bracket establishes non-ultralocality.
`eq:fact`; unnumbered $G(y)$; unnumbered good factorisation	Route B/C checks and interaction regimes; independent B for Toda/G blocks.	Pass. The good expression is the trigonometric continuation with its own S, D. No all-limits affine-Toda exclusion follows (R12).
Section 6 product-form exclusion (inline mixed derivative)	`smatrix_rejected_forms.py`.	The nonzero mixed derivative excludes $f(D) + g(S)$ in the tested bad rapidities. Narrow the summary tag or add the good check (R14).

Label audit beyond the displays. The explicit project formulas in Sections 2–6 are correctly classified as (c), subject to the corrections and test coverage just stated. The mutual-statistics interpretation, universality, cusp uniformisation, bound-state poles, resonance interpretation and exact S-matrix ideas are appropriately (d). The Toda/GHD and affine-Toda quantum proposals are appropriately (b). The missing-spin statement is algebraic and needs no unread textbook as its sole warrant. The opening assertion that the quantum Boussinesq theory "is" the W_3 charges should be read as describing that literature construction, not uniqueness of all possible quantisations. The broad stability, fusion-constant and affine-limit claims are the main places where the paper goes beyond its evidence. The first two already appear overstrong in older sources; copying them is not an independent justification **(c)**.

Phase 2i exclusion. The draft contains no Phase 2i Wiener–Hopf factors, edge coefficients, two-loop formula, corrected kernel or new physical synthesis in Sections 1–6, and no such result is imported into the remainder of the draft **(c)**. Its Section 8 placeholder names `toda_wh_and_kp_synthesis.pdf` solely as work awaiting audit. The quantum placeholders otherwise trace to the

earlier audits already cited in the master report. This is a status mention, not use of Phase 2i results. It can be removed if the PI intends a complete ban on even mentioning the filename; this review imports none of its claims.

Optional improvements

- Separate field, velocity and Toda sign notation: u_b, V, ϵ_T avoid the repeated use of v and s. **(d)**
- Give a short analytic derivation of the phase-gradient identity and of the real-root count. The numerical evidence is strong, but an explicit proof is more useful to a reader than a long collection of precisions. **(d)**
- Define scattering order once ($v_1 > v_2$) and distinguish $\delta(1,2)$ as an antisymmetric label function from the phase assigned to a specific physical incoming ordering. **(d)**
- Add a compact branch/endpoint convention for logarithms, the cusp, coincident solitons and breather boundary limits. For bad solitons say E increases with $|P|$, rather than saying $E(P)$ is monotone without specifying a direction branch. **(d)**
- Keep the quantum placeholders short. In particular, the identification of a retained linear phonon band with the full nonlinear CL invariant subspace is not independently proved by a finite phase integral; retain the heuristic qualification from `oneloop.tex`. **(d)**

Checks, reproducibility and failures recorded

Independent checks. Run `python3-Bpaper/scripts/review_draft1_checks.py`. The script writes no files, uses exact SymPy and 65-digit mpmath, asserts the new identities and explicit counterexamples, inventories citations, and ends with `ALLCHECKSPASSED`. Actual source function definitions are loaded via the Python AST without executing their top-level code or rewriting their result files. This makes the downstream comparison independent of saved JSON outputs. The optional `--run-sources` reruns the read-only scripts cited before Section 7, omitting the figure writer.

Original check requested by the PI. `paper/scripts/paper_conventions_check.py` was run unchanged and printed `ALLCHECKSPASSED`. Its checks establish the paper's energy and convention fixes, not the downstream dependency claims or every assertion in Sections 1–6.

Source-script coverage. The cited source scripts were read, and the read-only verification scripts for Sections 1–6 were rerun with bytecode writes disabled. The figure generator was inspected but not rerun, because it would overwrite an existing figure. Numerical root-search completeness and references not freshly read are not certified by these runs. The full Phase 2 one-loop and audit suites were not indiscriminately rerun: several write existing result files, contrary to the task constraint. The relevant actual functions were instead checked as described above. All 24 source-script processes exited successfully; 23 printed `ALLCHECKSPASSED`. The legacy `phase0/checks/convention_check.py` asserted its two symbolic residuals and printed the correct signs, but lacks the required success marker. Together with the separately run paper script, this covers the 25 verification scripts among the 26 cited paths (the remaining path is the figure generator). The independent review script also printed `ALLCHECKSPASSED`. **(c)**

Failures recorded (including repaired review checks).

1. The report's bad cubic sign fails the target Euler–Lagrange equation and positive-soliton energy. The associated positive Wick shift is also wrong for the corrected bad Hamiltonian. These are retained as explicit counterexamples, not repaired in the source files. **(c)**

2. The printed good energy fails direct quadrature and the velocity identity. The signed-root substitution at $\eta = -1$ fails nonlinear sewing; the middle transmission expression fails equality to z. The independent checks assert these nonzero residuals. **(c)**

3. The paper's phase-coordinate symplectic coefficient and unsigned left-moving Toda velocity fail direct checks. The unqualified strict stable-root count fails at a repeated-root threshold; the unsubtracted first hierarchy charge has a nonzero background density. **(c)**

4. In the first review-script run, the marginal-root test reused a SymPy symbol declared positive; the solver therefore excluded the desired root -1. A new real symbol fixes that test without changing its expected root or tolerance. **(c)**

5. An exploratory repository search used an invalid regular expression; it was replaced by narrower/fixed-string searches. One guessed script basename did not exist; the actual script inventory was then used. An initial full-text URL requested a nonexistent CZ version; version 1 was retrieved successfully. BC HTML and one PDF screenshot retrieval failed; its PDF text was available. The large CL HTML and the attempted Bona–Sachs DOI fetch were unavailable. These access failures limit fresh literature verification as stated above. **(c)**

6. Source verification contains limitations that must remain visible: the disabled `orTrue` assertion in the Hirota script, printed-only limit/cusp diagnostics, and the numerical stability search. No source assertion or tolerance was weakened by this review. **(c)**

7. The first PDF build failed because `xurl.sty` is unavailable; the document was changed to use the installed URL package's line-breaking hooks. A patch attempt with a stale context line was rejected without modifying files and was reapplied with the correct context. The legacy Phase 0 script's missing success marker is a reporting failure, not a failed mathematical residual. **(c)**

Verified external entries used in this review. Corrigan and Zambon, *Adding integrable defects to the Boussinesq equation*, arXiv:2306.03772; full-text relevant sections read. Bogdan and Charkina, *Quasiclassical quantization of the Boussinesq breather emerging from the kink localized mode*, arXiv:2502.14695; full-text relevant formula passages consulted. Charlier and Lenells, *On Boussinesq's equation for water waves*, arXiv:2204.02365; abstract/entry verified in this review, not a fresh full-text reading.